Journal of Ethnopharmacology 158 (2014) 102–112

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jep

Zuo-Gui and You-Gui pills, two traditional Chinese herbal formulas, downregulated the expression of NogoA, NgR, and RhoA in rats with experimental autoimmune encephalomyelitis

Shuang Kou [a,b,1], Qi Zheng [a,1], Yizhou Wang [a], Hui Zhao [a], Qiuxia Zhang [a], Ming Li [a], Fang Qi [a], Ling Fang [a], Lei Liu [a], Junyao Ouyang [a], Haiyu Zhao [c], Lei Wang [a,*]

[a] School of Traditional Chinese Medicine, Capital Medical University, Beijing 100069, PR China
[b] Institute of Basic Research in Clinical Medicine, China Academy of Chinese Medical Sciences, Beijing 100700, PR China
[c] Institute of Chinese Materia Medica, China Academy of Chinese Medical Sciences, Beijing 100700, PR China

ARTICLE INFO

Article history:
Received 5 May 2014
Received in revised form
12 September 2014
Accepted 8 October 2014
Available online 16 October 2014

Keywords:
Experimental autoimmune encephalomyelitis
Zuo-Gui pills
You-Gui pills
NogoA
Nogo receptor
RhoA

ABSTRACT

Ethnopharmacological relevance: Zuo-Gui pills (ZGPs) and You-Gui pills (YGPs) are 2 traditional Chinese herbal formulas used for treating multiple sclerosis (MS) in the clinical setting and have been shown to have neuroprotective effects in experimental autoimmune encephalomyelitis (EAE), an animal model of MS. The aim of this study was to explore the mechanisms underlying the neuroprotective functions of ZGPs and YGPs.

Materials and methods: Female Lewis rats were randomly divided into normal control, EAE model, 2 g/kg ZGP-treated EAE, 3 g/kg YGP-treated EAE, and prednisone acetate-treated groups. EAE model was induced by subcutaneous injection of MBP_{68-86} antigen. The neurological function scores were estimated. Histological structures of the brains and spinal cords were observed, and myelinated and axons imaged. NogoA, Nogo receptor (NgR), and RhoA transcript and protein levels were measured by real-time quantitative RT-PCR and western blotting on postimmunization (PI) days 14 (acute stage) and 28 (remission stage).

Results: ZGPs and YGPs significantly reduced neurological functions scores and abrogated inflammatory infiltrates, demyelination, and axonal damage. Furthermore, treatment with ZGPs and YGPs inhibited NogoA, NgR, and RhoA mRNA and protein expression in rats at both the acute and remission stages. ZGPs exhibited stronger effects on NogoA and RhoA expressions, as well as neurological function, during the acute stage of EAE, while YGPs caused greater reductions in NogoA expression during the remission stage.

Conclusions: Our findings suggested that ZGPs and YGPs exerted neuroprotective effects by down-regulation of NogoA, NgR, and RhoA pathways, with differences in response times and targets observed between ZGPs and YGPs.

1. Introduction

Multiple sclerosis (MS) is an inflammatory, demyelinating, neurodegenerative disease of the central nervous system (CNS) (Keegan and Noseworthy, 2002; Trapp and Nave, 2008; Heraud et al., 2010) and is the most frequent cause of neurological disability in young adults (Simmons et al., 2013). In recent years, an increasing number of reports have demonstrated the presence of axonal and neuronal damages in MS patients and experimental autoimmune encephalomyelitis (EAE), a commonly used animal model of MS, especially occurring during the early stages of the disease (Herz et al., 2010; Lassmann, 2010). Therefore, promoting nerve repair is an important strategy for the treatment of MS (Zhang et al., 2011). Basic research

Abbreviations: ZGP, Zuo-Gui pill; YGP, You-Gui pill; EAE, experimental autoimmune encephalomyelitis; MS, multiple sclerosis; CNS, central nervous system; TCM, traditional Chinese medicine; NgR, Nogo receptor; MBP, myelin basic protein; MTB, mycobacterium tubercusis; IFA, incomplete Freund's adjuvant; PI, postimmunization; HE, hematoxylin-eosin; TEM, transmission electron microscopy; qRT-PCR, quantitative reverse transcription polymerase chain reaction; PA, prednisone acetate; NC, normal control; MO, EAE model; PBS, phosphate-buffered saline; SDS-PAGE, sodium dodecyl sulfate polyacrylamide gel electrophoresis; GTP, guanosine triphosphatase; APP, amyloid precursor protein; GAP-43, growth associated protein-43; BDNF, brain-derived neurotrophic factor; cAMP/PKA, cyclic adenosine monophosphate/protein kinase A.

* Corresponding author. Tel.: +86 10 83911626; fax: +86 10 83911627.
E-mail address: tmwangl@ccmu.edu.cn (L. Wang).

[1] These authors contributed equally to this work.

http://dx.doi.org/10.1016/j.jep.2014.10.007
0378-8741/

has shown that lack of neurotrophic molecules, formation of glial scars, and inhibition of nerve regeneration may be important contributors to the development of MS/EAE. Inhibitory factors, such as myelin-associated glycoprotein (MAG), oligodendrocyte myelin glycoprotein (OMgp), NogoA (Cafferty et al., 2010), and other molecules involved in injury oligodendrocytes and myelination, exert their inhibitory effects via a common receptor, Nogo receptor (NgR), along with LINGO1 and either the neurotrophin receptor $p75^{NTR}$ or TROY (Fournier et al., 2001; Wang et al., 2002; Mi et al., 2004; Satoh et al., 2007). Inhibitory signals occur via activation of the small GTPase RhoA, which functionally results in growth cone collapse (Yamashita and Tohyama, 2003; Kubo and Yamashita, 2007). NogoA, NgR, and RhoA are strongly expressed in the lesions of MS and EAE (Karnezis et al., 2004; Satoh et al., 2005; Zhang et al., 2008), and expression of these factors has been shown to correlate with the clinical severity of EAE.

There has been no effective therapy for MS, especially in terms of neuroprotection. Novel therapies must therefore take into account the need for both immunomodulation and neuroprotection; as such, multifaceted treatment strategies are required (Van der Walt et al., 2010; Pifarre et al., 2014). Traditional Chinese medicine (TCM) has been shown to be effective in the treatment of MS. Using systematic reviews and meta-analyses, recent scholars have reported that Chinese herbal medicine could improve clinical symptoms, neurological signs, and immune functions and reduce the frequency of recurrence in MS patients (Liu et al., 2012). According to TCM theory, the basic pathogenesis of MS is deficiency of kidney essence, which was classified as kidney yin or yang deficiency clinically (Song and Fan, 2010). Therefore, 2 classic TCM formulas, named Zuo-Gui pills (ZGPs) and You-Gui pills (YGPs), first described by Jingyue Quanshu in the year 1624, are applied in MS treatment with nourishing kidney yin or warming kidney yang. In our previous studies, ZGPs and YGPs have been shown to reduce the clinical severity of MS (Fan et al., 2006; Song and Fan, 2010) and suppress ongoing EAE (Wang et al., 2008). Studies have indicated that ZGPs and YGPs have neuroprotective effects and may be associated with brain-derived neurotrophic factor (BDNF) and cyclic adenosine monophosphate/protein kinase A (cAMP/PKA) pathways (Wang et al., 2014). However, whether these formulas have regulatory roles in nerve regeneration by inhibiting specific signaling pathways is still unknown.

In the present study, we observed the effects of ZGPs and YGPs on NogoA, NgR, and RhoA inhibitory pathways in EAE rats. We also compared the similarities and differences between these 2 formulas.

2. Materials and methods

2.1. Reagents

Prednisone acetate (PA) was purchased from Tianjin Pacific Pharmaceutical Co. Ltd. (China). Myelin basic protein (MBP) $_{68-86}$ (YGSLPQKSQRSQDENPV) was synthesized by Beijing SciLight Biotechnology Co. Ltd. (China), and the purity was greater than 95%. Incomplete Freund's adjuvant (IFA) and mycobacterium tuberculosis (MTB) were purchased from Difco Co. Ltd. (USA). NogoA rabbit anti-rat antibodies were from Abcam Co. Ltd. (UK), NgR rabbit anti-rat antibodies were from Millipore Co. Ltd. (USA). RhoA rabbit anti-rat antibodies were from Cell Signaling Technology Co. Ltd. (USA). DEPC H_2O and TRIzol reagent were from Invitrogen Co. Ltd. (USA). The Fermentas K1622 RT reverse transcription ELISA kit was from MBI Co. Ltd. (USA). The SYBR Green PCR Master Mix kit was from American Application Biology Co. Ltd. (USA).

2.2. Preparation of ZGPs and YGPs

ZGPs and YGPs were produced by Henan Wanxi Pharmaceutical Co. Ltd. with batch numbers as 090701 and 090403. Voucher specimens were deposited at TCM brain disease laboratory of School of Traditional Chinese Medicine, Capital Medical University, China. ZGPs and YGPs were composed of 8 and 10 kinds of herbal medicines respectively (Tables 1 and 2). They were prepared in consistent with the standards of Chinese Ministry of Public Health. Briefly, both of them were obtained by grinding Chinese medicines into fine powder, sifted and well mixed with Honey. To ensure the quality and stability of ZGPs and YGPs, the ultra-high performance liquid chromatography combined with LTQ-Orbitrap mass spectrometer (UHPLC-Orbitrap MS) was used to identify the active ingredients.

2.3. UHPLC-Orbitrap MS analysis of ZGPs and YGPs

The analysis of the extract was performed by UHPLC-Orbitrap MS with Thermo BDS HYPERSIL C18 column (2.1 mm × 150 mm, ID 2.4 μm). The column temperature was set at 35 °C. The mobile phase was water (0.1% formic acid, phase A) and acetonitrile (phase B) with a gradient program as follows: 0.0–30.0 min, 99–88% A; 30.0–32.0 min, 88–85% A; 32.0–55.0 min, 85–75% A; 55.0–60.0 min, 75–60% A; 60.0–70.0 min, 60–0% A. The flow rate was 0.3 ml/min. All solvents were filtered through a 0.45 μm filter before use. The injection volume was 3 μL. The MS spectra were acquired in positive ion mode with an electrospray ionization source (ESI+). The ESI source condition was ionspray voltage, 5000 V; nebulizer, 35 psi; capillary temperature, 350 °C; capillary voltage, 35 V; the resolution was set at 30,000. Full-scan spectra were acquired in the mass range m/z 50–1000. The identified compounds of ZGPs and YGPs were shown in Figs. 1 and 2, respectively. The high resolution MS data of betaine, sweroside, loganin, hyperin, quercitrin, chlorogenic acid, songorine, fuziline, neoline and cinnamaldehyde is listed in Tables 3 and 4.

2.4. Quantification of loganin in ZGPs and YGPs

According to the standard of Chinese Pharmacopeias (2010 Edition), the contents of loganin in ZGPs and YGPs were quantitatively

Table 1
Chinese medicines contained in ZGPs.

Chinese name	Latin name	Family	Part used
Shu di huang	Rehmanniae radix praeparata	*Rehmannia glutinosa* (Gaertn.) DC.	Root
Shan yao	Dioscoreae rhizoma	*Dioscorea oppositifolia* L.	Rootstock
Gou qi zi	Lycii fructus	*Lycium barbarum* L.	Fruit
Shan zhu yu	Corni fructus	*Cornus officinalis* Siebold & Zucc.	Fruit
Chuan niu xi	Cyathulae radix	*Cyathula officinalis* K.C.Kuan	Root
Lu jiao jiao	Cervi cornus colla	*Cervus elaphus* Linnaeus	horn
Gui jia jiao	Testudinis carapacis et plastri Colla	*Chinemys reevesii* (Gray)	Shell
Tu si zi	Cuscutae semen	*Cuscuta chinensis* Lam.	Seed

Note: The ratio of these herbs was 8:4:4:4:3:4:4:4.

Table 2
Chinese medicines contained in YGPs.

Chinese name	Latin name	Family	Part used
Shu di huang	Rehmanniae radix praeparata	*Rehmannia glutinosa* (Gaertn.) DC.	Root
Shan yao	Dioscoreae rhizoma	*Dioscorea oppositifolia* L.	Rootstock
Shan zhu yu	Corni fructus	*Cornus officinalis* Siebold & Zucc.	Fruit
Gou qi zi	Lycii fructus	*Lycium barbarum* L.	Fruit
Tu si zi	Cuscutae semen	*Cuscuta chinensis* Lam.	Fruit
Lu jiao jiao	Cervi cornus colla	*Cervus elaphus* Linnaeus	horn
Du zhong	Eucommiae cortex	*Eucommia ulmoides* Oliv.	Bark
Rou gui	Cinnamomi cortex	*Cinnamomum cassia* (L.) J.Presl	Bark
Dang gui	Angelicae sinensis radix	*Angelica sinensis* (Oliv.) Diels	Root
Fu zi	Aconiti lateralis radix preparata	*Aconitum carmichaelii* Debeaux	Root

Note: The ratio of these herbs was 8:4:3:4:4:4:4:2:3:2.

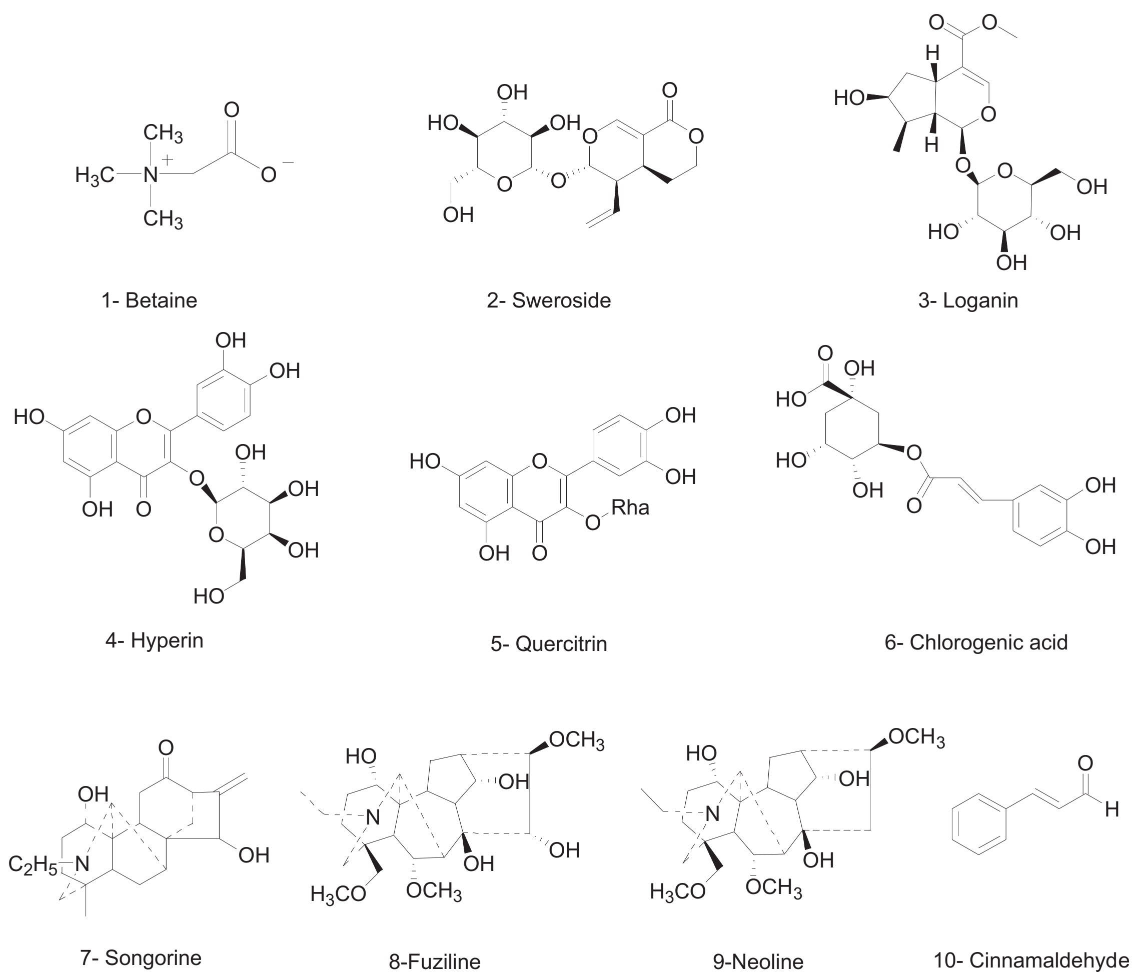

Fig. 1. Chemical structures of active compounds from ZGPs and YGPs.

analyzed. The pills were ground into fine powder first. The powder was ultra-sonicated (300 W, 40 kHz) with 50% methanol for 15 min followed with a heated reflux for 1 h. The samples were analyzed by HPLC. The contents of loganin in ZGPs and YGPs were 0.33 mg/ml and 0.34 mg/ml, respectively.

2.5. Animals

SPF-grade female Lewis rats weighing 150–180 g were obtained from Beijing Vital River Laboratories (China). Rats were housed in the pathogen-free animal facility of Experimental Animal Center of Capital Medical University, Beijing, China. All the animal procedures were approved by the Ethics Committee of Capital Medical University.

2.6. Treatments

Rats were randomly divided into 5 groups: normal control (NC), EAE model (MO), PA-treated EAE, ZGP-treated EAE, and YGP-treated EAE groups ($n=30$ rats each). The rats in the ZGP- and YGP-treated EAE groups were treated with suspensions of 2 g/kg ZGP and 3 g/kg YGP, respectively, after immunization, as described below. The rats in the PA-treated EAE group were treated with 6 mg/kg PA when the clinical signs appeared. The rats in NC and MO groups were treated with normal saline. All of the rats were treated once per day for 28 days. The dose administered was determined according to the following formula: D_R (dose per kg body weight)$=D_H \times R_R/R_H \times (W_H/W_R)^{1/3}$, as detailed in The

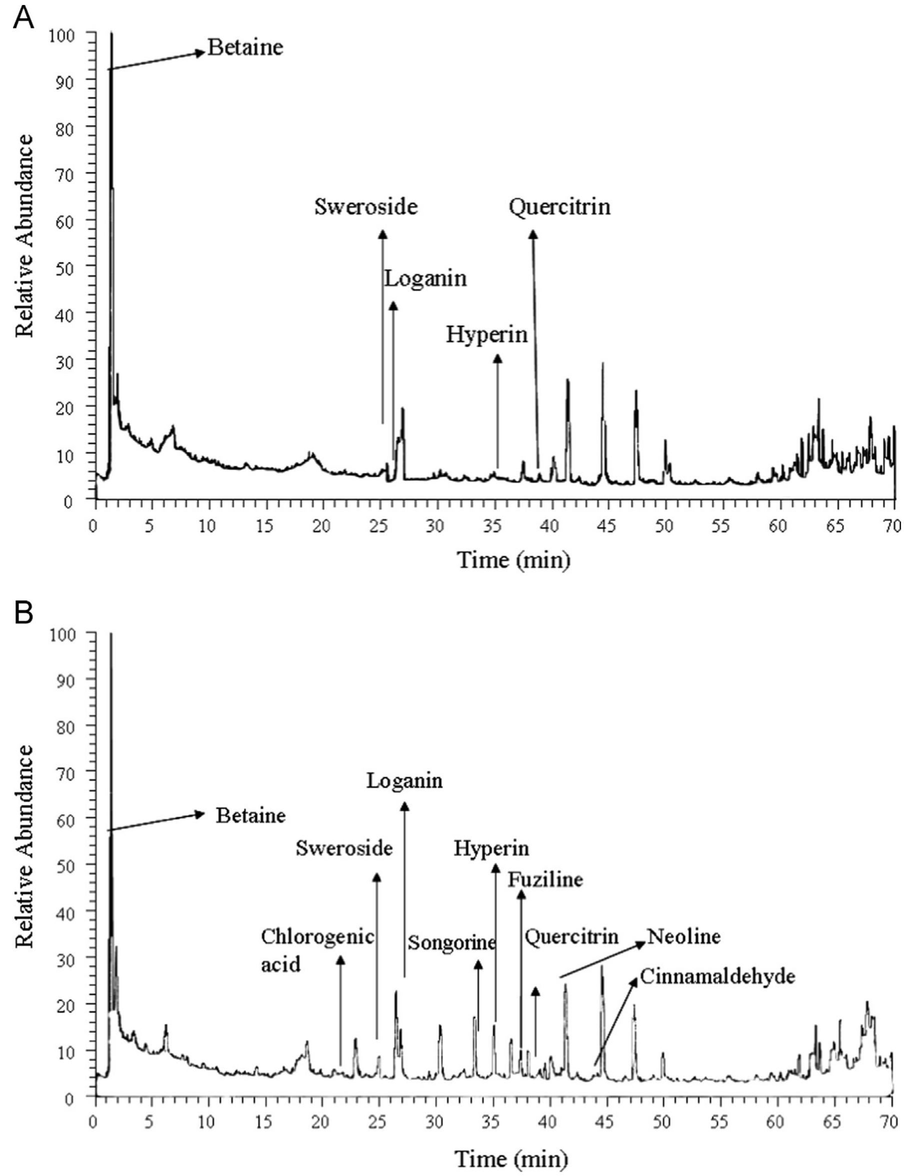

Fig. 2. HPLC chromatogram of ZGPs (A) and YGPs (B).

Table 3
Components of ZGPs identified by HPLC-MS.

Peak no.	Retention time (min)	Selected ion	Formula	Compound	Measured mass (m/z)	Diff (ppm)	Source of drugs
1	1.34	$[M+H]^+$	$C_5H_{11}O_2N$	Betaine	118.08611	2.666	Lycii fructus
2	24.94	$[M+H]^+$	$C_{16}H_{22}O_9$	Sweroside	359.13425	1.646	Corni fructus
3	26.45	$[M+H]^+$	$C_{17}H_{26}O_{10}$	Loganin	391.16049	1.346	Corni fructus
4	35.02	$[M+H]^+$	$C_{21}H_{20}O_{12}$	Hyperin	465.10324	1.048	Cuscutae semen
5	38.74	$[M+H]^+$	$C_{21}H_{20}O_{11}$	Quercitrin	449.10867	1.853	Cuscutae semen, Eucommiae cortex

Methodology of Pharmacological Experiment. D_H and D_R are doses for humans and rats, and W_H and W_R are body weights of humans and rats, respectively. R_H and R_R are the coefficients of 0.1 for humans and 0.09 for rats, respectively (Xu et al., 2005).

2.7. EAE model

For induction of EAE, 200 μL antigen containing 50 μg MBP_{68-86} in 100 μL phosphate-buffered saline (PBS), 100 μL IFA, and 2 mg MTB were injected subcutaneously into 2 hind footpads of rats, as previously described (Mao et al., 2007).

2.8. Sample collection

Rats were sacrificed at different time points (days 14, acute stage; or days 28, remission stage). The acute stage was defined as the day at which peak disease was observed by neurological function scores, while the remission stage was the day at which the signs of EAE were no longer worsening. Four rats from each group were used for hematoxylin-eosin (HE) staining. The rats were anesthetized with 10% chloral hydrate (350 mg/kg body weight, intraperitoneally), the chest was opened, and the heart was exposed. A catheter was inserted into the left ventricle, and the right atrial appendage was

Table 4
Components of YGPs identified by HPLC-MS.

Peak no.	Retention time (min)	Selected ion	Formula	Compound	Measured mass (m/z)	Diff (ppm)	Source of Chinese medicine
1	1.34	$[M+H]^+$	$C_5H_{11}O_2N$	Betaine	118.08610	−0.155	Lycii fructus
2	24.94	$[M+H]^+$	$C_{16}H_{22}O_9$	Sweroside	359.13440	0.741	Corni fructus
3	26.45	$[M+H]^+$	$C_{17}H_{26}O_{10}$	Loganin	391.16058	0.707	Corni fructus
4	35.02	$[M+H]^+$	$C_{21}H_{20}O_{12}$	Hyperin	465.10373	0.978	Cuscutae semen
5	38.74	$[M+H]^+$	$C_{21}H_{20}O_{11}$	Quercitrin	449.10834	0.502	Cuscutae semen, Eucommiae cortex
6	21.48	$[M+H]^+$	$C_{16}H_{18}O_9$	Chlorogenic acid	355.10233	−0.029	Eucommiae cortex
7	33.50	$[M+H]^+$	$C_{22}H_{31}O_3N$	Songorine	358.23828	0.610	Aconiti lateralis radix preparata
8	37.35	$[M+H]^+$	$C_{24}H_{39}O_7N$	Fuziline	454.28125	1.321	Aconiti lateralis radix preparata
9	40.95	$[M+H]^+$	$C_{24}H_{39}O_6N$	Neoline	438.28616	1.146	Aconiti lateralis radix preparata
10	43.83	$[M+H]^+$	C_9H_8O	Cinnamaldehyde	133.06500	0.209	Cinnamomi cortex

cut. Rats were then perfused immediately with 200 mL normal saline by a peristaltic pump, followed by a slow infusion of 200 mL 4% paraformaldehyde. The brains and spinal cords of rats were removed and fixed in the same perfusion buffer for a week. One rat from each group was randomly chosen for observation by transmission electron microscopy (TEM). The rats were perfused with 200 mL PBS, followed by 2% paraformaldehyde and 2% glutaraldehyde. One-millimeter cross-sections of the cerebral white matter and lumbar region of the spinal cord were removed and fixed in 3% glutaraldehyde overnight, following 3 rinses in PBS. Sections were then stored in 0.1 mol/L phosphate buffer at 4 °C. Brains and spinal cords of rats were quickly removed, immediately frozen in liquid nitrogen, and stored at −80 °C until analysis of mRNA with qRT-PCR and protein with western blotting.

2.9. Neurological function scores in rats

After the day of immunization (day 0), the neurological functions scores of EAE rats were observed daily. A quantitative neurological scale of 15-point was used, based on previous reports (Weaver et al., 2005; Dasilva and Yong, 2008). For the tail: 0, no signs; 1, half-paralyzed tail; 2, fully paralyzed tail. For each limb: 0, no signs; 1, weak or altered gait; 2, paresis; 3, fully paralyzed limb.

2.10. Histopathology and TEM observations

Histopathological observations in brains and spinal cords of rats were evaluated by light microscopy following HE staining. The tissues of rats were embedded in paraffin. Sections (thickness, 5 μm) were stained with HE and observed with a light microscope (Nikon Eclipse 80i). The Inflammatory cell infiltration was scored as follows: 0, no infiltrates; 1, sporadic inflammatory cells; 2, single inflammatory cells surrounding blood vessels; 3, large amount of inflammatory cell infiltration surrounding blood vessels and 4, inflammatory cell infiltration and perivascular cuff formation, or diffuse infiltration of inflammatory cells without apparent perivascular cuff formed (Zappia et al., 2005).

TEM was used to observe changes in the ultrastructures of axons and myelination in the brains and spinal cords of rats. The specimens were postfixed in 2% osmium tetroxide in 0.1 mol/L cacodylate buffer, then dehydrated in a graded series of ethanol (50–100%), transferred to propylene oxide, and embedded in Epon. Epon blocks were sectioned on a Leica EM ultramicrotome. The sections were stained with 1% toluidine blue, and ultrathin sections were stained with uranyl acetate and lead citrate. The sections were viewed on a TEM (JEM-1230, Japan) using a digital camera system for micrographs.

2.11. Real-time quantitative RT-PCR

Total RNA was isolated from frozen brains and spinal cords of rats using TRIzol reagent. The quality of extracted RNA was checked by agarose gel electrophoresis, and cDNA synthesis was performed using a Fermentas K1622 RT reverse transcription kit, according to the manufacturer's instructions. Primers were designed for RT-PCR using GenBank Accession sequences and were synthesized by Shanghai Sangon Biotechnology Co., Ltd. GAPDH mRNA served as an internal control. The sequences of primers were as follows: NogoA: forward, 5′-GGTGCCTTGTTCAATGGTCT-3′, reverse, 5′-AATC-TGCTTTGCGCTTCAAT-3′; NgR: forward, 5′-CACGGCACATCAA-TGAC-TCT-3′, reverse, 5′-TGCGGTTCTTTCTGGAACA-3′; RhoA: forward, 5′-TATTGAAGTG-GACGGGAAGC-3′, reverse, 5′-GGGCACATTTGGACA-GAAAT-3′; and GAPDH: forward, 5′-CCATGGAGAAGGCTGGG-3′, reverse, 5′-CAAAGTTGTCATGGATGACC-3′. The sample reaction volume was 12 μL, containing 2 μg RNA, 1 μL Oligo DT, and the remaining volume of DEPC water. The real-time reaction mixtures contained 10 μL REAL SYBR Mixture (2×), 0.5 μL of the upstream primer (10 μmol/L), 0.5 μL of the downstream primer (10 μmol/L), 1.5 μl of template, and 7.5 μL DEPC water, for a final reaction volume of 20 μL. The reaction conditions were set according to the manufacturer's protocol for the SYBR Green PCR Master Mix kit; after the reaction, a final extension step was carried out at 72 °C for 10 min, and PCR products were stored at 4 °C. Data were generated using an Applied Biosystem 7500 Real-Time PCR System and analyzed using the $^{\Delta\Delta}Ct$ method.

2.12. Western blotting

Cerebral white matter and spinal cords of rats were removed from the freezer, and proteins were extracted after homogenization with protein extraction agent (50 mmoL/L Tris–HCl, pH 7.4; 10% SDS; 1 mmoL/L PMSF, 0.25% deoxidationcholic acid sodium) for 30 min on ice. The samples were then centrifuged at 12,000 rpm at 4 °C for 10 min, supernatants were collected, and protein concentrations were measured by BCA assay. Protein bands were separated by SDS-polyacrylamide gel electrophoresis (PAGE), and subsequently transferred onto nitrocellulose membranes (Millipore). Membranes were probed with primary anti-NogoA (1:5000), anti-NgR (1:5000), anti-RhoA (1:1000), and anti-GAPDH (1:10,000) antibodies, followed by horseradish peroxidase-labeled anti-rabbit IgG (1:12,000) as the secondary antibody. The membranes were then incubated in electrochemiluminescent reagent for 5 min, followed by exposure to Kodak film and development with an enhanced chemiluminescent kit. Images were scanned using theYLN-2111 gel image analysis system (Yalien, China) and processed using Image J (National Institutes of Health, USA). The integrated absorbance was quantified using Image Quant TL 2005 software.

2.13. Statistics analysis

All the values were expressed in means ± SEMs. SPSS 13.0 was used to analyze data with the single-factor variance analysis method, and comparisons among groups were carried out using the LSD test. For abnormally distributed data, the rank sum test was used. Differences with p-values of less than 0.05 were considered significant.

3. Results

3.1. Treatment with ZGPs and YGPs decreased neurological function scores of EAE rats

Rats showed signs of disease by PI days 8, exhibiting increasingly low spirits, weight loss, feeble and loose hanging tails, and lack of strength/paralysis of the hind limbs. The peak neurological function score was reached on PI days 13. Scores decreased in treated EAE rats at different time points. In both ZGP- and YGP-treated EAE rats, scores decreased on PI days 10, 16, and 19–21, as compared with MO rats ($P<0.05$ or $P<0.01$). In addition, scores reduced only in ZGP-treated EAE rats on PI days 12, 17, 18, and 22–24 as compared with MO rats ($P<0.05$), scores reduced in PA-treated EAE rats on PI days 10, 16 and 20–24 as compared with MO rats ($P<0.05$ or $P<0.01$), scores reduced in YGP-treated EAE rats on PI days 16 and in ZGP-treated EAE rats on PI days 17 as compared with PA-treated rats ($P<0.05$). The decrease of the scores in PA-treated EAE rats was better than that of ZGP- and YGP-treated EAE rats on PI days 26 ($P<0.05$, Table 5).

3.2. ZGPs and YGPs eliminated pathological changes in EAE rats

A large number of inflammatory cells were aggregated around small blood vessels in the EAE group on PI days 14. Conversely, inflammatory cell infiltration was dramatically reduced by treatment with PA, ZGP, or YGP. Pathological changes in the brains and spinal cords of rats on PI days 28 were similar to those on PI days 14 (Figure is not shown). The scores of inflammatory cells infiltration in the MO, PA-, ZGP- and YGP-treated groups increased significantly on PI days 14 and 28, as compared to the NC group ($P<0.01$, respectively). However, the scores were decreased significantly following treatment with PA, ZGP, or YGP in EAE rats, as compared to MO rats ($P<0.01$ or $P<0.05$, respectively) except YGP-treated group on PI days 14 in spinal cords (Fig. 3).

3.3. ZGPs and YGPs attenuated axon and myelin damage and promoted repair of axons/myelin

We also studied the structures of axons and the myelin sheath using TEM analysis on PI days 14. In the EAE model group, the structure of myelin sheath layers in white matter and spinal cords of rats was loose, fractured or lost, or indistinct and fused, and a granular substance was separated from the tissue. Pathological axilemma peripheral clearance appeared between the myelin sheath and axon. Axons were atrophic or swollen, microtubules and microfilaments were fractured, mitochondria were swollen, and axonal ridges had merged and disappeared. Many necrotic neurons were observed, with accompanying inflammatory cell infiltration. Changes in the ultrastructures of brains and spinal cords in PA-, ZGP-, and YGP-treated EAE rats were milder than those in EAE model rats; damaged myelin sheaths, axons, and organelles were restored to different extents following the different treatments (Fig. 4).

3.4. ZGPs and YGPs decreased the expression of NogoA, NgR, and RhoA mRNA in EAE rats

qRT-PCR showed that NogoA mRNA in brains of rats in the MO group increased on PI days 14 and 28, with a significant increase observed on PI days 14, as compared to the NC group ($P<0.05$). In contrast, NogoA mRNA was significantly decreased on PI days 14 following treatment with PA, ZGP, or YGP in EAE rats, as compared to MO rats ($P<0.01$ or $P<0.05$, respectively). On PI days 28, only YGP-treated EAE rats exhibited significantly decreased NogoA mRNA, as compared to MO rats ($P<0.05$). In spinal cords of rats, NogoA mRNA increased significantly in MO rats on PI days 14 and 28, as compared to NC rats ($P<0.01$, $P<0.05$, respectively), but decreased obviously in PA-, ZGP-, and YGP-treated EAE rats, as compared to MO rats on PI days 14 ($P<0.05$ or $P<0.01$). However, there were no significant changes among PA-, ZGP-, and YGP-treated EAE rats on PI days 28.

In brains of rats, NgR mRNA was significantly increased in MO rats as compared to NC rats on PI days 28 ($P<0.01$). NgR mRNA was significantly increased in PA- and ZGP-treated EAE rats as compared to NC rats on PI days 14 ($P<0.05$ or $P<0.01$, respectively), but was significantly decreased in YGP-treated EAE rats as compared to ZGP-treated EAE rats on PI days 14 ($P<0.01$). In addition, NgR mRNA decreased significantly in PA-treated EAE rats on days 28 PI, as compared to MO rats ($P<0.05$). In spinal cords of rats, there were no significant differences in NgR mRNA between the 5 groups on PI days 14. However, on PI days 28, NgR mRNA increased significantly in MO rats compared to NC rats ($P<0.01$), but decreased significantly in PA- and ZGP-treated EAE rats ($P<0.05$ or $P<0.01$, respectively).

There were no significant differences in RhoA mRNA in brains between the 5 groups on PI days 14 and 28. However, in spinal

Table 5
Effects of ZGPs and YGPs on neurological function scores in EAE rats at different time points.

Group	Days after immunization										
	8	9	10	11	12	13	14	15	16	17	18
MO	0.11 ± 0.11	0.67 ± 0.47	3.44 ± 0.96	4.44 ± 1.38	10.78 ± 1.23	11.33 ± 0.60	9.22 ± 0.89	7.13 ± 0.30	6.13 ± 0.13	3.75 ± 0.45	3.50 ± 0.50
PA	0.10 ± 0.10	0.20 ± 0.20	1.10 ± 0.71*	4.00 ± 1.03	9.50 ± 1.07	10.40 ± 0.75	8.20 ± 0.53	6.40 ± 0.22	4.60 ± 0.43**	3.70 ± 0.37	2.60 ± 0.31
ZGP	0.00 ± 0.00	0.00 ± 0.00	0.10 ± 0.10**	3.00 ± 0.70	7.30 ± 0.97*	9.50 ± 0.67	8.60 ± 0.40	6.90 ± 0.28	3.80 ± 0.33**	2.40 ± 0.27*#	2.40 ± 0.27*
YGP	0.00 ± 0.00	0.20 ± 0.13	0.50 ± 0.34**	5.80 ± 1.50	9.90 ± 0.90	10.40 ± 0.75	8.50 ± 0.75	6.40 ± 0.22	3.60 ± 0.27**#	2.80 ± 0.33	2.60 ± 0.31
Group	Days after immunization										
	19	20	21	22	23	24	25	26	27	28	
MO	3.25 ± 0.53	2.88 ± 0.58	2.50 ± 0.60	2.25 ± 0.65	2.00 ± 0.71	1.63 ± 0.65	1.25 ± 0.73	0.88 ± 0.52	1.00 ± 0.46	0.63 ± 0.26	
PA	2.40 ± 0.27	1.90 ± 0.10*	1.20 ± 0.13**	1.10 ± 0.18*	0.60 ± 0.22*	0.50 ± 0.17*	0.30 ± 0.15	0.20 ± 0.13	0.30 ± 0.15	0.40 ± 0.16	
ZGP	2.20 ± 0.20*	1.40 ± 0.34**	1.20 ± 0.20**	0.90 ± 0.23*	0.90 ± 0.23*	0.60 ± 0.22*	1.00 ± 0.15	1.00 ± 0.15#	0.80 ± 0.20	0.80 ± 0.13	
YGP	2.20 ± 0.20*	1.80 ± 0.13*	1.60 ± 0.16*	1.30 ± 0.26	1.10 ± 0.23	0.80 ± 0.20	1.00 ± 0.00	0.90 ± 0.10#	0.90 ± 0.10	0.80 ± 0.20	

Data are expressed as the mean ± SEM.

* $P<0.05$.

** $P<0.01$ vs. MO group.

# $P<0.05$ vs. PA group.

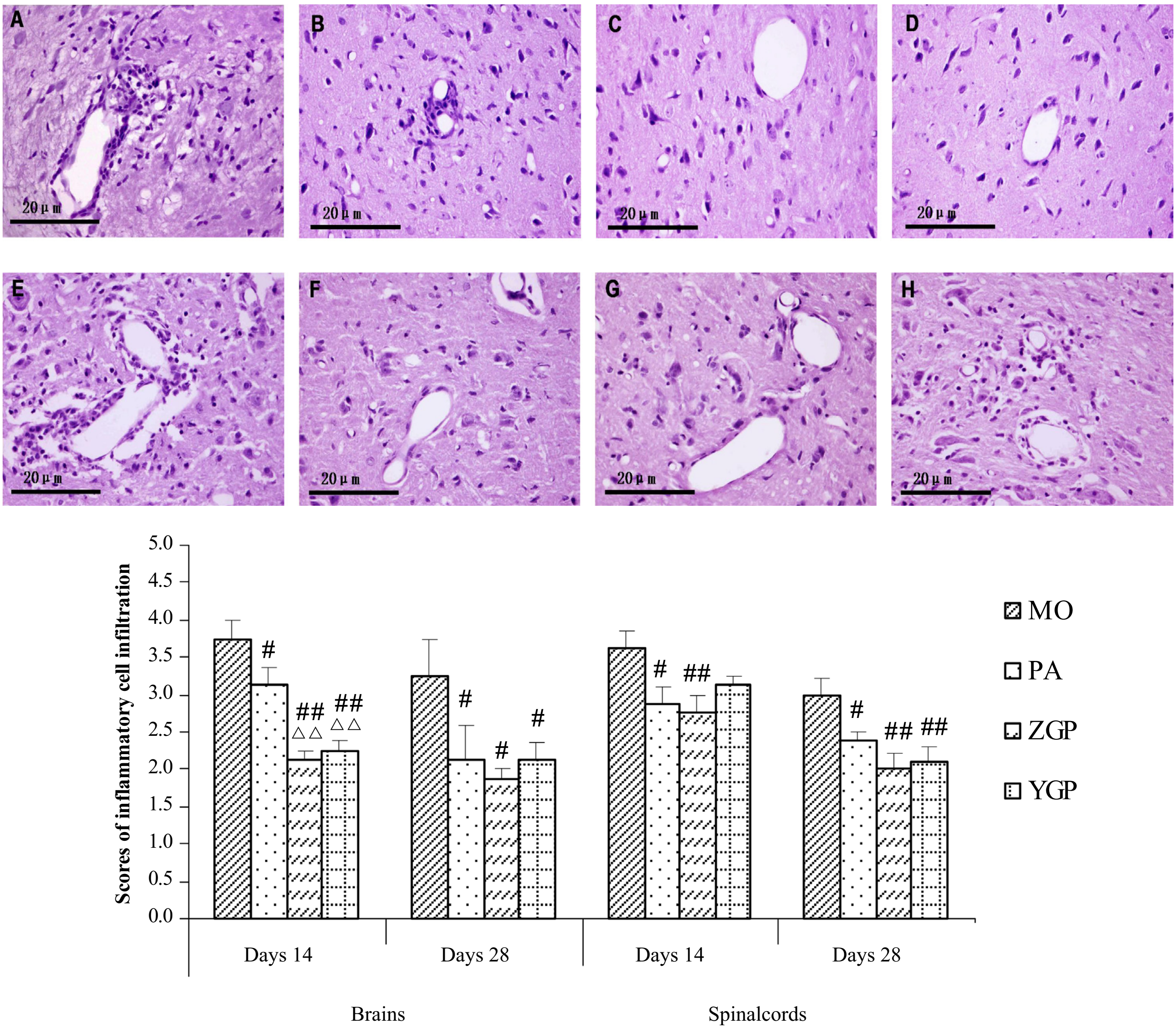

Fig. 3. Histological changes in the brains and spinal cords of rats under light microscopy with HE staining on PI days 14 and 28. (A) to (E) and (F) to (H) show the histological changes in the brains and spinal cords of rats in the EAE, PA-, ZGP- and YGP-treated groups on PI days 14, respectively. Scale bars: magnification: 400 ×, $n=4$ for each group. The Inflammatory cell infiltration was assessed by the scores. $^{\#}P<0.05$, $^{\#\#}P<0.01$ vs. MO; $^{\triangle\triangle}P<0.01$ vs. PA.

cords, RhoA mRNA increased significantly in MO rats as compared to the NC rats on PI days 14 and 28 ($P<0.05$), but decreased significantly in YGP-treated EAE rats on PI days 14 and in PA-, ZGP-, and YGP-treated EAE rats on PI days 28, as compared to MO rats ($P<0.05$ or $P<0.01$, respectively; Fig. 5).

3.5. ZGPs and YGPs decreased expression of NogoA, NgR, and RhoA protein expression in EAE rats

Western blots showed that NogoA protein increased significantly in the brains of MO rats compared to NC rats on PI days 14 ($P<0.05$). NogoA protein was also increased in MO rats on PI days 28, but there was no statistical significance. However, NogoA protein was significantly decreased in PA-, ZGP-, and YGP-treated EAE rats on PI days 14 and 28, as compared to MO rats ($P<0.05$). NogoA protein did not exhibit an obvious increase in the spinal cords of MO rats as compared to NC rats on PI days 14 and 28, but decreased obviously in PA-, ZGP-, and YGP-treated EAE rats as compared to MO rats on PI days 14 ($P<0.01$ or $P<0.05$, respectively) and in YGP-treated EAE rats on PI days 28 ($P<0.05$).

NgR protein increased significantly in the brains of MO rats as compared to NC rats on PI days 14 ($P<0.05$). In addition, while NgR increased on PI days 28, there was no statistical significance. NgR protein was also decreased significantly in PA-, ZGP-, and YGP-treated EAE rats as compared to MO rats on PI days 14 and 28 ($P<0.05$ or $P<0.01$, respectively). NgR protein did not exhibit obvious differences in the spinal cords in MO rats as compared to NC rats on PI days 14 and 28, but decreased obviously in PA- and ZGP-treated EAE rats as compared to MO rats on PI days 14 ($P<0.01$, $P<0.05$, respectively). There were no significant differences in NgR protein among the 5 groups on PI days 28.

RhoA protein increased significantly in the brains of MO rats as compared to NC rats on PI days 14 and 28 ($P<0.05$), but decreased significantly in ZGP- and YGP-treated EAE rats on PI days 14 and in PA-, ZGP-, and YGP-treated EAE rats on PI days 28, as compared to MO rats ($P<0.01$ or $P<0.05$, respectively). RhoA protein increased

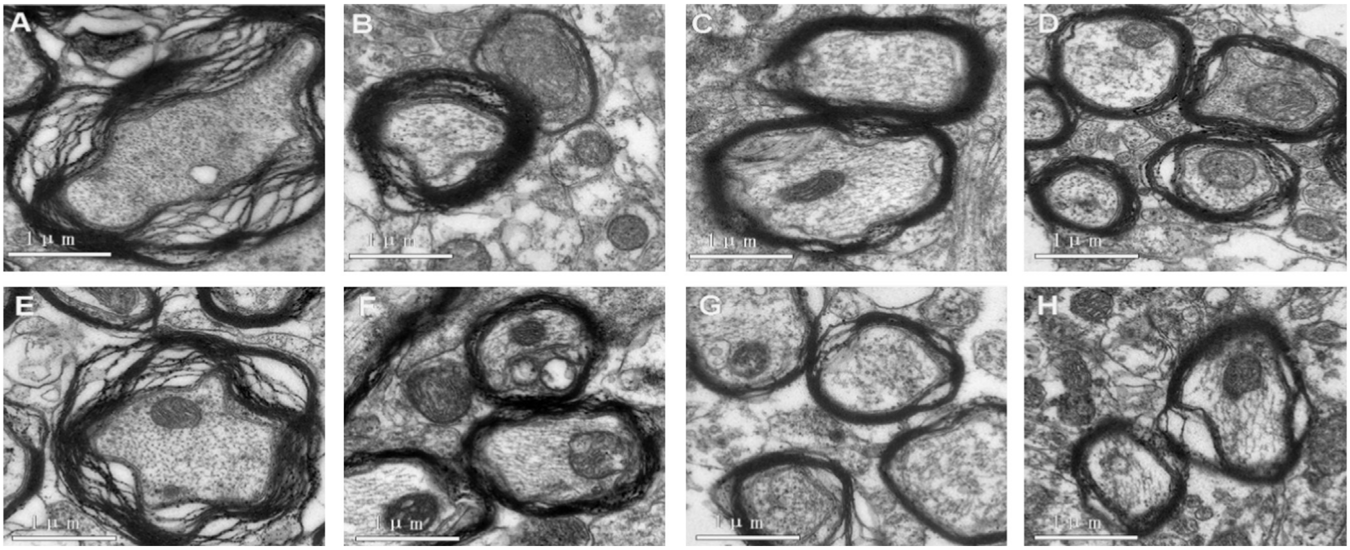

Fig. 4. Ultrastructural changes in the axons and myelin sheaths of the brains and spinal cords of rats under TEM on PI days 14. (A) and (E) Brains and spinal cords of rats in the EAE group; (B) and (F) brains and spinal cords of rats in the PA-treated group; (C) and (G) brains and spinal cords of rats in the ZGP-treated group; (D) and (H) brains and spinal cords of rats in the YGP-treated group. Scale bars: magnification: 50,000×, $n=1$ for each group.

significantly in the spinal cords of MO rats as compared to NC rats on PI days 14 and 28 ($P<0.01$), but decreased significantly in PA-, ZGP-, and YGP-treated EAE rats as compared to MO rats ($P<0.01$, Fig. 6).

4. Discussion

The above results suggested that ZGPs and YGPs exerted neuroprotective effects by downregulating NogoA, NgR, and RhoA pathways. In addition, we observed some differences between ZGPs and YGPs in terms of timing, affected site, and targets.

In this study, we induced EAE by injecting MBP antigen. The features of EAE were confirmed by the observed marked increase in neurological function scores, the existence of inflammatory infiltrates in brains and spinal cords of EAE rats, and damage to axons and the myelin sheath. While treatment with ZGPs and YGPs permitted the recovery of severity scores and histopathological changes, myelin and axon damages exhibited even greater recovery following treatment with ZGPs or YGPs. These results indicated that ZGPs and YGPs exerted neuroprotective effects, with ZGPs having stronger effects than YGPs in the recovery of neurological function scores.

We further observed the mechanisms through which ZGPs and YGPs affected the mRNA and protein expression of inhibitory molecules, such as NogoA, NgR, and RhoA, in the acute stage and remission stage of EAE. A large number of studies have indicated that NogoA, NgR, and RhoA are important factors in autoimmune demyelinating diseases, such as MS and EAE (Teng and Tang, 2005; Fontoura and Steinman, 2006). Inflammatory lesions of these diseases have been shown to result in demyelination and axonal damage, and inhibition of the NogoA–NgR–RhoA signaling pathways prevents new growth cone formation (Satoh et al., 2005; Zhang et al., 2008). In addition, increased NogoA and NgR mRNA and protein have been observed in the chronic phase of EAE (Theotokis et al., 2012). In the study, accumulation of RhoA (+) cells was maximal in EAE rat brains during the acute stage, and this upregulation was sustained until the recovery stage of the disease. RhoA expression has been shown to be spatially associated with MS lesions, with expression levels exhibiting a marked decrease from active lesions during chronic stages (Zhang et al., 2008). Furthermore, immunization of animals with Nogo-66 induced EAE similar to the classic EAE model (Fontoura et al., 2004). Our data also showed similar results, indicating that NogoA, NgR, and RhoA mRNA and protein increased in the brains and/or spinal cords of rats during the acute and/or remission stages. In contrast, NogoA-knockout or -knockdown mice developed a milder disease course of EAE (Karnezis et al., 2004; Yang et al., 2010), and RhoA inhibitors have been shown to attenuate the progression of EAE (Gao et al., 2013). However, despite all of these advances in MS research, the neuroprotective treatment of MS is still a major challenge in the clinical setting.

Chinese herbal medicines play an important role in the prevention and treatment of neural diseases, and recently, some reports have described interventions targeting NogoA/NgR with Chinese medicine (Qin et al., 2012). However, few studies have investigated the effects of treatment during different stages of EAE. We analyzed the effects of ZGPs and YGPs on NogoA, NgR, and RhoA mRNA and protein expression during the course of EAE in Lewis rats. During the acute stage, both ZGPs and YGPs dramatically downregulated the mRNA expression of NogoA in brains and spinal cords, indicating the major role of ZGP in the regulation of NogoA in the brains. In addition, YGPs dramatically downregulated RhoA mRNA in the spinal cord. In addition, NgR mRNA was highly upregulated in ZGP-treated EAE rats, although the differences in NgR mRNA expression may indicate more refined reactions within the CNS, and such reactions require further detailed studies. Regarding the protein expression of inhibitory factors, our data showed that both ZGPs and YGPs dramatically downregulated NogoA and RhoA proteins in brains and spinal cords and NgR protein in brains only, indicating that ZGPs had a critical role in the regulation of NogoA, NgR, and RhoA in the CNS. During the remission stage, both ZGPs and YGPs dramatically downregulated RhoA mRNA in the spinal cord, indicating that ZGPs played a major role in the regulation of RhoA. In addition, ZGPs and YGPs exerted dramatic effects on the mRNA and protein expression of the other factors in the brains and/or spinal cords during this stage as well, with YGP exerting particularly dramatic effects on the regulation of NogoA in the spinal cords. As a control agent, PA exhibited similar functions.

Our research group has focused on the prevention and treatment of MS and EAE for 10 years. In previous studies, treatment of MBP-induced EAE rats with ZGPs and YGPs was shown to alleviate neurological disability, inhibit cellular apoptosis, regulate cellular immune factors, and reduce inflammatory infiltrates, demyelination, and axonal damage. It also promotes axonal regeneration by regulating MBP, neurofilament-200, amyloid precursor protein

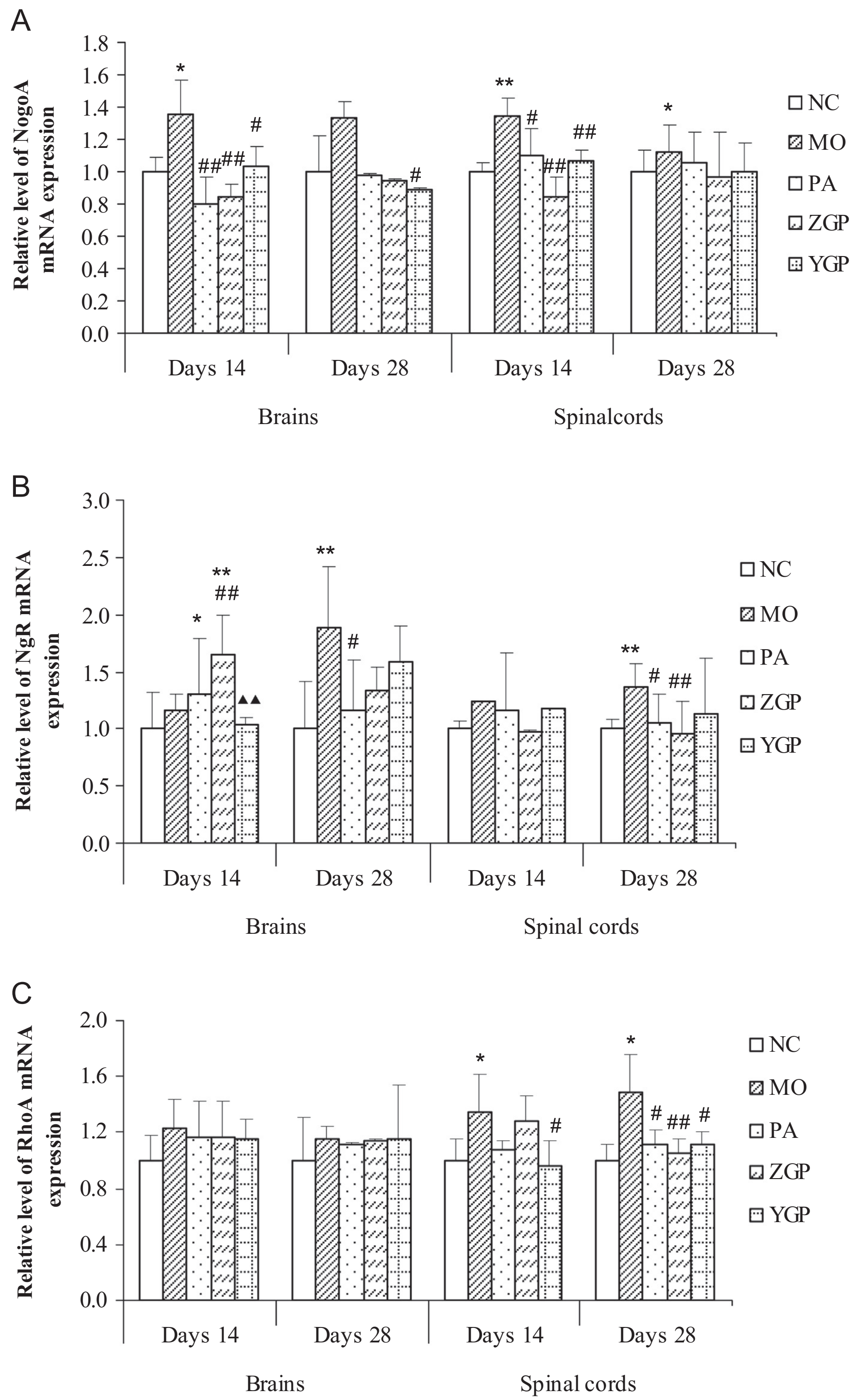

Fig. 5. Effects of ZGPs and YGPs on the expression of NogoA, NgR, and RhoA mRNA in brains and spinal cords of rats on days 14 and 28. (A) qRT-PCR of NogoA mRNA. (B) qRT-PCR of NgR mRNA. (C) qRT-PCR of RhoA mRNA. Data are represented as the mean ± SEM, $n=5$ for each group. $^{*}P<0.05$, $^{**}P<0.01$ vs. NC; $^{\#}P<0.05$, $^{\#\#}P<0.01$ vs. MO. $^{▲▲}P<0.01$ vs. ZGPs.

(APP), and growth-associated protein-43; therefore, ZGPs and YGPs played a role in neuroprotection in EAE animals (Wang et al., 2008, 2010, 2011; Kou et al., 2011). Our studies also found differences between the effects of ZGPs and YGPs, similar to the effects observed in other studies. In a very recent study, ZGPs were shown to play a more prominent role in regulating APP mRNA expression in EAE rat brains during the acute stage, while YGPs significantly improved BDNF mRNA expression of EAE animal brains and spinal cords during the remission stage (Wang et al., 2014). In our study, both ZGPs and YGPs inhibited the expression of NogoA, NgR, and RhoA mRNA and/or protein during the acute or remission stage to different extents. ZGPs played a more prominent role in decreasing neurological function scores and NogoA and RhoA expression during the acute stage of EAE, while YGPs were more important for regulation of NogoA during the remission stage. The above results suggested that there were differences between ZGPs with nourishing yin and YGPs with warming yang over time, in different tissues, and of different targets.

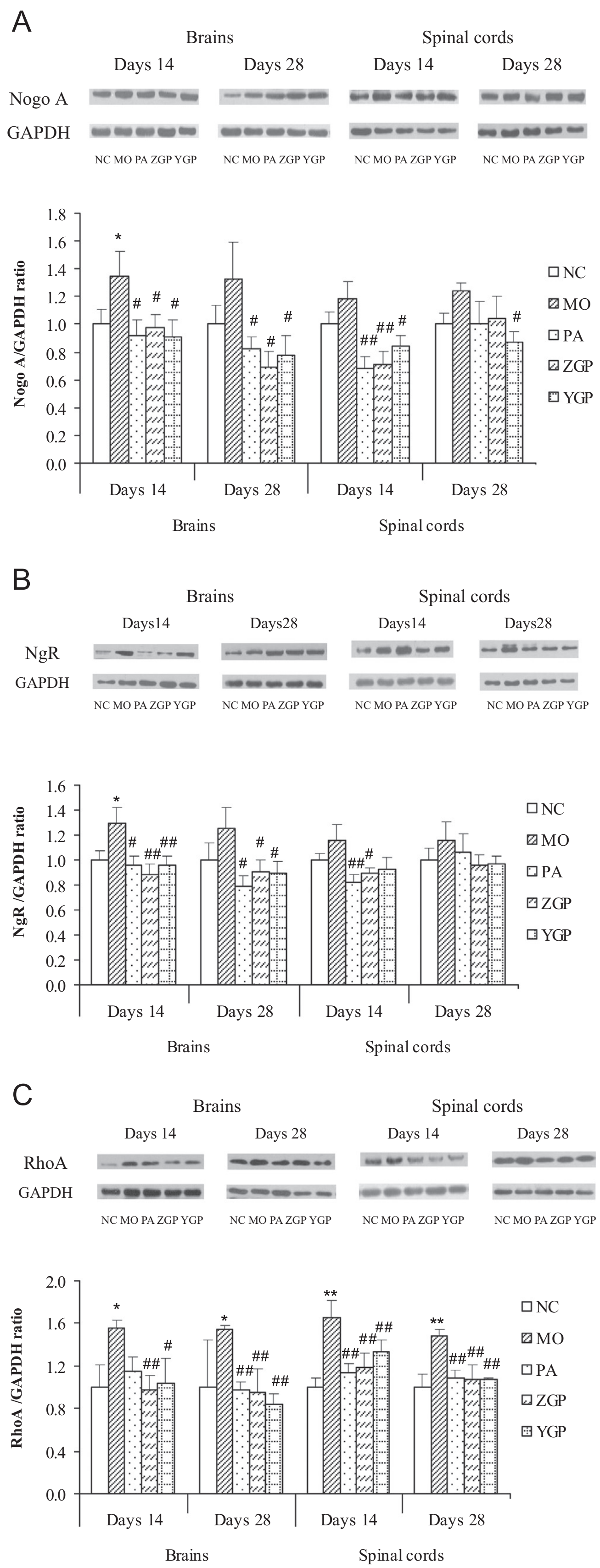

Fig. 6. Effects of ZGPs and YGPs on the expression of NogoA, NgR, and RhoA protein in brains and spinal cords of rats on days 14 and 28. (A) Western blots of NogoA protein. (B) Western blots of NgR protein. (C) Western blots of RhoA protein. Data are represented as the mean ± SEM, $n=5$ for each group. $^{*}P<0.05$, $^{**}P<0.01$ vs. NC; $^{\#}P<0.05$, $^{\#\#}P<0.01$ vs. MO.

ZGPs and YGPs may be useful multifunctional compounds for prevention of permanent functional deficits during the treatment of MS patients. Indeed, we have recently shown that these 2 formulas promoted the recovery of the neurological function of the paralyzed limbs by both limiting inflammation and promoting damaged nerve repair. The mechanism most likely involved a reduction of the inhibitory effects on axons through downregulation of NogoA, NgR, and RhoA levels and upregulation of neurotrophic factors (i.e., NGF and BDNF) to improve axonal regeneration of the microenvironment, promoting nerve repair. These data provide insights into the development of novel treatment strategies for MS. With regard to the apparent selective effects of these 2 formulas in the regulation of nerve regeneration, our data suggested that ZGPs and YGPs had roles in tonifying kidneys, enriching essences, nourishing yin, and warming yang through the observed molecular pathways.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (30873230 and 81273742), Beijing Natural Science Foundation (7092014 and 7132031), and Program for Changcheng Scholars of the Importation and Development of High-Caliber Talents Project of Beijing Municipal Institutions (CIT&TCD20140329).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.jep.2014.10.007.

References

Cafferty, W.B., Duffy, P., Huebner, E., Strittmatter, S.M., 2010. MAG and OMgp synergize with Nogo-A to restrict axonal growth and neurological recovery after spinal cord trauma. Journal of Neuroscience 30, 6825–6837.

Dasilva, A.G., Yong, V.W., 2008. Expression and regulation of matrix metalloproteinase-12 in experimental autoimmune encephalomyelitis and by bone marrow derived macrophages in vitro. Journal of Neuroimmunology 199, 24–34.

Fan, Y.P., Wang, P., Zhang, X.H., Gong, H.Y., Zhou, L., Liu, X.Z., Wang, L., 2006. Treatment of relapsing multiple sclerosis with Erhuang Formula. Journal of Beijing University on Traditional Chinese Medicine 29 (273–276), 280.

Fontoura, P., Ho, P.P., DeVoss, J., Zheng, B., Lee, B.J., Kidd, B.A., Garren, H., Sobel, R.A., Robinson, W.H., Tessier-Lavigne, M., Steinman, L., 2004. Immunity to the extracellular domain of Nogo-A modulates experimental autoimmune encephalomyelitis. Journal of Immunology 173, 6981–6992.

Fontoura, P., Steinman, L., 2006. Nogo in multiple sclerosis: growing roles of a growth inhibitor. Journal of the Neurological Sciences 245, 201–210.

Fournier, A.E., GrandPre, T., Strittmatter, S.M., 2001. Identification of a receptor mediating Nogo-66 inhibition of axonal regeneration. Nature 409, 341–346.

Gao, C., Huang, L., Long, Y., Zheng, J., Yang, J., Pu, S., Xie, L., 2013. Y-39983, a selective Rho-kinase inhibitor, attenuates experimental autoimmune encephalomyelitis via inhibition of demyelination. Neuroimmunomodulation 20, 334–340.

Heraud, P., Caine, S., Campanale, N., Karnezis, T., McNaughton, D., Wood, B.R., Tobin, M.J., Bernard, C.C., 2010. Early detection of the chemical changes occurring during the induction and prevention of autoimmune-mediated demyelination detected by FT-IR imaging. Neuroimage 49, 1180–1189.

Herz, J., Zipp, F., Siffrin, V., 2010. Neurodegeneration in autoimmune CNS inflammation. Experimental Neurology 225, 9–17.

Karnezis, T., Mandemakers, W., McQualter, J.L., Zheng, B., Ho, P.P., Jordan, K.A., Murray, B.M., Barres, B., Tessier-Lavigne, M., Bernard, C.C., 2004. The neurite outgrowth inhibitor Nogo A is involved in autoimmune-mediated demyelination. Nature Neuroscience 7, 736–744.

Keegan, B.M., Noseworthy, J.H., 2002. Multiple sclerosis. Annual Review of Medicine 53, 285–302.

Kou, S., Wang, Y.Z., Li, M., Q.F., Zheng, Q., Zhao, H., Wang, L., 2011. Effects of Zuogui Pill and Yougui Pill on expression of apoptosis protein fas and bax of brain tissue in rat with experimental autoimmune encephalomyelitis. Journal of Liaoning University on Tradittion Medicine 13, 61–65.

Kubo, T., Yamashita, T., 2007. Rho-ROCK inhibitors for the treatment of CNS injury. Recent Patents on CNS Drug Discovery 2, 173–179.

Lassmann, H., 2010. Axonal and neuronal pathology in multiple sclerosis: what have we learnt from animal models. Experimental Neurology 225, 2–8.

Liu, J., Gao, Y., Kan, B.H., Zhou, L., 2012. Systematic review and meta-analysis of randomized controlled trials of Chinese herbal medicine in treatment of multiple sclerosis. Journal of Chinese Integrative Medicine 10, 141–153.

Mao, Y.S., Lu, C.Z., Wang, X., Xiao, B.G., 2007. Induction of experimental autoimmune encephalomyelitis in Lewis rats by a viral peptide with limited homology to myelin basic protein. Experimental Neurology 206, 231–239.
Mi, S., Lee, X., Shao, Z., Thill, G., Ji, B., Relton, J., Levesque, M., Allaire, N., Perrin, S., Sands, B., Crowell, T., Cate, R.L., McCoy, J.M., Pepinsky, R.B., 2004. LINGO-1 is a component of the Nogo-66 receptor/p75 signaling complex. Nature Neuroscience 7, 221–228.
Pifarre, P., Gutierrez-Mecinas, M., Prado, J., Usero, L., Roura-Mir, C., Giralt, M., Hidalgo, J., Garcia, A., 2014. Phosphodiesterase 5 inhibition at disease onset prevents experimental autoimmune encephalomyelitis progression through immunoregulatory and neuroprotective actions. Experimental Neurology 251, 58–71.
Qin, X.D., Kang, L.Y., Liu, Y., Huang, Y., Wang, S., Zhu, J.Q., 2012. Chinese medicine's intervention effect on Nogo-A/NgR. Evidence-Based Complementary and Alternative Medicine 2012, 528482. http://dx.doi.org/10.1155/2012/528482 (ID).
Satoh, J., Onoue, H., Arima, K., Yamamura, T., 2005. Nogo-A and nogo receptor expression in demyelinating lesions of multiple sclerosis. Journal of Neuropathology and Experimental Neurology 64, 129–138.
Satoh, J., Tabunoki, H., Yamamura, T., Arima, K., Konno, H., 2007. TROY and LINGO-1 expression in astrocytes and macrophages/microglia in multiple sclerosis lesions. Neuropathology and Applied Neurobiology 33, 99–107.
Simmons, S.B., Pierson, E.R., Lee, S.Y., Goverman, J.M., 2013. Modeling the heterogeneity of multiple sclerosis in animals. Trends in Immunology 34, 410–422.
Song, L.J., Fan, Y.P., 2010. Influence of treatment based on syndrome differentiation mainly with tonifying the kidney on cytokines in plasm of multiple sclerosis at acute stage. China Journal of Traditional Chinese Medicine and Pharmacy 25, 745–748.
Teng, F.Y., Tang, B.L., 2005. Nogo signaling and non-physical injury-induced nervous system pathology. Journal of Neuroscience Research. 79, 273–278.
Theotokis, P., Lourbopoulos, A., Touloumi, O., Lagoudaki, R., Kofidou, E., Nousiopoulou, E., Poulatsidou, K.N., Kesidou, E., Tascos, N., Spandou, E., Grigoriadis, N., 2012. Time course and spatial profile of Nogo-A expression in experimental autoimmune encephalomyelitis in C57BL/6 mice. Journal of Neuropathology and Experimental Neurology 71, 907–920.
Trapp, B.D., Nave, K.A., 2008. Multiple sclerosis: an immune or neurodegenerative disorder. Annual Review of Neuroscience 31, 247–269.
Van der Walt, A, Butzkueven, H., Kolbe, S., Marriott, M., Alexandrou, E., Gresle, M., Egan, G., Kilpatrick, T., 2010. Neuroprotection in multiple sclerosis: a therapeutic challenge for the next decade. Pharmacology and Therapeutics 126, 82–93.
Wang, K.C., Kim, J.A., Sivasankaran, R., Segal, R., He, Z., 2002. P75 interacts with the Nogo receptor as a co-receptor for Nogo, MAG and OMgp. Nature 420, 74–78.
Wang, L., Zhao, H., Fan, Y.P., Gong, H.Y., Li, M., Qi, F., Liu, Y., 2010. Research on the mechanism of Zuogui Pill and Yougui Pill in promoting axonal regeneration in model rats of autoimmune encephalomyelitis. Journal of Chinese Integrative Medicine 16, 167–172.
Wang, Y.Z., Kou, S., Gu, L.Y., Zheng, Q., Li, M., Qi, F., Zhao, H., Wang, L., 2014. Effects of Zuogui Pill and Yougui Pill on the expression of brain-derived neurotrophic factor and cyclic adenosine monophosphate/protein kinase a signaling transduction pathways of axonal regeneration in model rats with experimental autoimmune encephalomyelitis. Journal of Chinese Integrative Medicine 20, 24–30.
Wang, L., Fan, Y.P., Gong, H.Y., Ye, M., Zhou, L., Zhao, H., Mu, Y., Yan, X.Y., 2008. Effect of Zuogui Pill and Yougui Pill on remyelination and axonal regeneration in rats with experimental allergic encephalomyelitis. China Journal of Experimental Traditional Medical Formulae. 4, 42–45.
Wang, Y.Z., Kou, S., Yan, C.C., Zheng, Q., Zhang, Q., Zhao, H., Wang, L., 2011. Effects of Zuogui Pill and Yougui Pill on the expression of APP and GAP-43. Chinese Journal of Integrative Medicine on Cardio-/Cerebrovascular Disease 9, 1221–1224.
Weaver, A., Goncalves, D.S.A., Nuttall, R.K., Edwards, D.R., Shapiro, S.D., Rivest, S., Yong, V.W., 2005. An elevated matrix metalloproteinase (MMP) in an animal model of multiple sclerosis is protective by affecting Th1/Th2 polarization. The FASEB Journal 19, 1668–1670.
Xu, S.Y., Bian, R.L., Chen, X., 2005. Methodology of Pharmacological Experiment, 3rd People's Medicinal Publishing House, Beijing, pp. 202–204.
Yamashita, T., Tohyama, M., 2003. The p75 receptor acts as a displacement factor that releases Rho from Rho-GDI. Nature Neuroscience 6, 461–467.
Yang, Y., Liu, Y., Wei, P., Peng, H., Winger, R., Hussain, R.Z., Ben, L.H., Cravens, P.D., Gocke, A.R., Puttaparthi, K., Racke, M.K., McTigue, D.M., Lovett-Racke, A.E., 2010. Silencing Nogo-A promotes functional recovery in demyelinating disease. Annals of Neurology 67, 498–507.
Zappia, E., Casazza, S., Pedemonte, E., Benvenuto, F., Bonanni, I., Gerdoni, E., Giunti, D., Ceravolo, A., Cazzanti, F., Frassoni, F., Mancardi, G., Uccelli, A., 2005. Mesenchymal stem cells ameliorate experimental autoimmune encephalomyelitis inducing T-cell anergy. Blood 106, 1755–1761.
Zhang, J., Kramer, E.G., Asp, L., Dutta, D.J., Navrazhina, K., Pham, T., Mariani, J.N., Argaw, A.T., Melendez-Vasquez, C.V., John, G.R., 2011. Promoting myelin repair and return of function in multiple sclerosis. FEBS Letters 585, 3813–3820.
Zhang, Z., Schittenhelm, J., Meyermann, R., Schluesener, H.J., 2008. Lesional accumulation of RhoA(+) cells in brains of experimental autoimmune encephalomyelitis and multiple sclerosis. Neuropathology and Applied Neurobiology 34, 231–240.